Adwaita Game Library — Main Spec

Nov 4, 2026 · @Tim

Summary and goals

We build a native GNOME app (GTK4 + libadwaita, shipped as a Flatpak on Flathub) that gathers every PC game on the machine into one library laid out like the Xbox library section, plus a fullscreen mode for controllers. The desktop look and behaviour borrow from Cartridges, Heroic and Faugus; the fullscreen mode borrows from OpenGamepadUI. Working title: TBD.

Goals

- **G1 One library.** Import games from Steam, Heroic (Epic, GOG, Amazon), Lutris, Bottles, Faugus, desktop entries and manual entries into one list.
- **G2 Xbox-style browsing.** Filters, sort, a cover grid in three tile sizes, collections and a recents row, built from stock libadwaita and GTK widgets.
- **G3 Launch anything.** Hand off to the owning launcher where one exists; run standalone Windows games through umu-launcher and Proton, Faugus-style.
- **G4 Couch mode.** A fullscreen, fully controller-navigable 10-foot interface over the same library.
- **G5 Flathub quality.** A Flatpak that passes the Flathub linter and quality guidelines, with every permission justified.

Non-goals for v1

- No store front: buying, downloading and updating store games stays in Steam and Heroic.
- Not a session or OS shell: no gamescope session, TDP or hardware control (OpenGamepadUI's territory).
- No emulation front end.
- No Xbox name, logo or brand green: we copy the layout, not the brand.
- No accounts or cloud sync.

Users and design principles

The app serves Linux players whose games are spread over several launchers and who want one GNOME-native place to find and start them, at a desk or on a couch.

User	Situation	What they need from us
Desktop player	GNOME desktop, games in Steam, Heroic and loose .exe folders	One searchable library, fast launch, native look
Couch or handheld player	TV, HTPC or handheld with a controller	Fullscreen mode navigable without keyboard or mouse
Tinkerer	Standalone Windows games, per-game Proton tweaks	Add a game by hand, pick Proton version and tools

Core use cases

1. Find a game by name, source or collection and start it in two actions or fewer.
2. See at a glance what is installed, what is not, and what was played last.
3. Group games into user collections and pin favourites.
4. Switch to fullscreen and browse, launch and return using only a controller.
5. Add a standalone Windows game, set its Proton version and launch it.

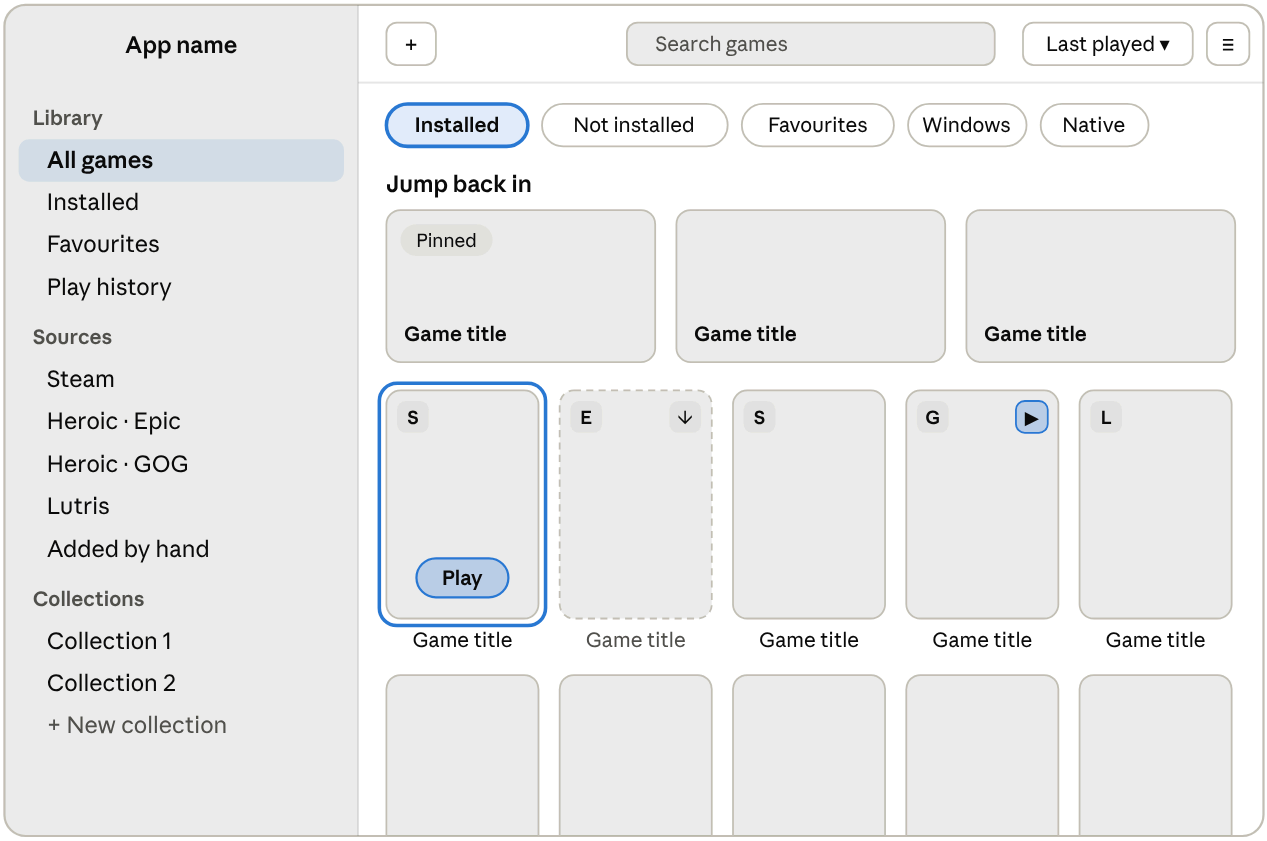
Principles

- **Adwaita first.** Stock libadwaita widgets, system accent colour and typography; no custom theme. Layout comes from Xbox, look and behaviour from GNOME.
- **Launchers stay owners.** We read other launchers' data and hand launches back to them; we never write into their files.
- **One model, two shells.** Desktop and fullscreen modes share the same data models and services; only the presentation differs.
- **Offline first.** The library works without network; artwork and metadata are fetched and cached when online.
- **Honest sandbox.** Ask only for permissions a feature needs, explain each in the README and MetaInfo, and degrade gracefully when one is missing.

Desktop UI: navigation, library and game page

The main window is a split view: a sidebar of library views, sources and collections beside a content pane with search, sort, stackable filters, a "Jump back in" row and a 2:3 cover grid; activating a game pushes a full-page game hub with Play focused.

Sources and collections on the left; filters, recents and covers on the right



Launcher badge top left, state badge top right (↓ not installed, ▶ running); Play shows on hover or focus

Library window wireframe · sidebar, header, filters, Jump back in, cover grid

The focused tile shows the accent ring and Play pill; a dashed cover is a game that is not installed. Art, titles and collection names are placeholders.

Window structure

- `AdwApplicationWindow` → `AdwToastOverlay` → `AdwOverlaySplitView` (sidebar | content), with an AdwSidebar (libadwaita 1.9+) for the rows. The HIG advises against fixed sidebars in immersive content apps, so the sidebar can be hidden with F9 or a header toggle. The content side is an `AdwNavigationView`: library page, then game page.

- The sidebar collapses at a breakpoint around 730 sp (as in Cartridges' rewrite); the layout must work down to 360 sp wide.

Sidebar

Section	Rows	Behaviour
Library	All games, Installed, Favourites, Play history	Count per row; Play history lists games grouped by Today, This week, Earlier
Sources	One row per detected launcher (Steam, Heroic: Epic/GOG/Amazon, Lutris, Bottles, Faugus, Desktop, Added by hand) with icon and count	Rows with zero games hidden; a source can be hidden entirely in Preferences
Collections	User collections, then a "New collection" row	Rename, reorder, delete from the row's context menu

Header bar (content side)

- Start: sidebar toggle (always shown; F9), Add menu (Add game, Import now).
- Centre: search entry; typing anywhere in the library starts a search.
- End: sort menu, main menu (Fullscreen mode, Preferences, Keyboard shortcuts, About).
- Sort options: Last played (default), Name A–Z, Name Z–A, Recently added, Playtime.

Library page

1. Filter row: toggle buttons that stack, Xbox-mode style: Installed, Not installed, Favourites, Windows (Proton), Native; a Clear action when any is on.
2. "Jump back in" row: up to six recently played games as landscape hero cards, pinned games first (up to three pins); hidden while a search or filter is active.
3. Cover grid: `GtkGridView` of 2:3 covers at 200×300 logical size (tile size Small, Medium, Large in Preferences), title below (can be turned off), launcher badge top left, state badge top right (not installed, running, needs permission). Badges stay out of the bottom third of the art.
4. A Play button appears on hover and keyboard focus; activating the tile opens the game page.
5. Context menu (right-click, long-press, Menu key): Play, Details, Favourite, Add to collection, Pin to Jump back in, Edit, Hide, Open in launcher, Remove (with undo).

6. Empty states with `AdwStatusPage`: no games (Import button), no search results, empty collection, nothing hidden.

Game page

- Hero art across the top with a gradient and the logo (or title) over it; below, a pill `.suggested-action` Play button with default focus, then Favourite, Add to collection and a ⋮ menu (Edit, Hide, Open in launcher, Open install folder, Remove).
- Facts row: launcher, playtime, last played, date added, install size when known.
- Preference groups below: About (developer, genres, description when the source has them); Launch (for games we run: Proton version, prefix, arguments, environment, MangoHud, GameMode; for store games a read-only summary and "Open in Heroic" or "Open in Steam").

Dialogs and feedback

- Add or Edit game: `AdwDialog` with cover, hero and logo pickers (file, URL, SteamGridDB), and rows for title, executable, arguments, working folder, type (Native or Windows), Proton version, prefix.
- Preferences (`AdwPreferencesDialog`): General (tile size, titles, badges, exit after launch), Sources (enable, location override, hide), Artwork (SteamGridDB key, prefer SteamGridDB), Windows games (default Proton, prefix folder, tools), Fullscreen mode.
- Toasts with Undo for hide, remove and collection changes; import progress as a small header indicator, never a modal.

Keyboard and desktop integration

- Ctrl+F search, Ctrl+N add game, Ctrl+, preferences, F9 sidebar, Ctrl+H show hidden, Enter play or open, Alt+Left back, F11 toggles fullscreen mode, Ctrl+? shortcuts; Delete only acts on the focused tile and always with undo.
- Every tile has an accessible label (title, launcher, state).
- A GNOME Shell search provider that queries the running app's model over D-Bus, and a `--launch <id>` command line for desktop shortcuts.

Research: Research — Xbox Game Library UI · Research — Cartridges · Research — Heroic Games Launcher · Research — Faugus Launcher

Game sources, importers and launching

v1 reads other launchers' data read-only and hands store games back to their own launcher through URI handlers; games with no launcher of their own (desktop entries, Flatpak games, Faugus games, native and Windows games added by hand) are started by us, through host spawn or umu inside the sandbox, and tracked for playtime.

Sources

Source	Detection (native, then Flatpak)	Data we read	Launch route	Playtime	Milestone
Steam	`~/.steam/steam`, `xdg-data/Steam`, `~/.var/app/com.valvesoftware.Steam/data/Steam`	`libraryfolders.vdf` (every library; its `apps` list where a library sits outside granted paths), `appmanifest_*.acf` (installed when `StateFlags & 4`), `appinfo.vdf` (type, platform, developer), `librarycache` art	Store URI: `steam://rungameid/<id>`, or `steam://launch/<id>/dialog` per game	Imported from Steam (`localconfig.vdf` per account, to verify)	M1
Heroic: Epic, GOG, Amazon, sideload	`~/.config/heroic`, `~/.var/app/com.heroicgameslauncher.hgl/config/heroic`	`store_cache/*_library.json` joined with each `installed.json`; `store/game-overrides.json`; hidden, favourites, categories from `store/config.json`	Store URI: `heroic://launch?appName=…&runner=…`	Imported (`store/timestamp.json`)	M1
Lutris	`~/.local/share/lutris`, `~/.var/app/net.lutris.Lutris/data/lutris`	`pga.db`, copied then queried; `coverart/`	Store URI: `lutris:rungameid/<id>`	Imported (`pga.db`, column to verify)	M2
Bottles	`xdg-data/bottles`, `~/.var/app/com.usebottles.bottles/data/bottles`	`library.yml`, `data.yml`	Store URI: `bottles:run/<bottle>/<game>` (used by Cartridges)	–	M2
Faugus	`xdg-data/faugus-launcher`, `~/.var/app/io.github.Faugus.faugus-launcher/data/faugus-launcher`	`games.json`, `covers/`, `banners/`	Host spawn of `faugus-launcher --game <id>`	Imported (`games.json`), not tracked by us	M2
Desktop entries and Flatpak games	`xdg-data/applications`, `/run/host/usr/share/applications`, Flatpak `exports` dirs	`.desktop` files with `Categories=Game`, skipping known launchers' own entries and their game shortcuts	Host spawn of the parsed Exec line, or `flatpak run <id>`	Own sessions	M2
Manual native game	Game folder chosen through the portal	Executable, title, arguments; art from SteamGridDB	Host spawn (M2), or Steam Linux Runtime via umu (M3)	Own sessions	M2
Manual Windows game	Game folder chosen through the portal	Title and icon from the .exe; art from SteamGridDB	umu and Proton inside the sandbox	Own sessions	M3
itch, standalone Legendary	Paths as in Cartridges	`butler.db`, `installed.json`	`itch://caves/<id>/launch`, `legendary launch`	–	Backlog

Import rules

- Never write to another launcher's files and never read credential files (for example Heroic's `gog_store/auth.json`).
- Each source has a location resolver: ordered native and Flatpak candidates, required sub-paths, first match and any user override stored in SourceState.
- Game IDs are `<source>:<native id>` and stable. A game gone from its source gets `missing_since` and leaves the views, returning with its user data if it reappears; a game the user removes is marked user-removed and stays removed while it is still in the source.
- Sync runs at start-up, on `Gio.FileMonitor` changes, and on manual refresh, off the main thread with a small progress indicator in the header.
- Import failures are per entry and collected into one summary dialog, never a modal per error.
- Hidden, favourite and category flags from a source are applied once, at first import; after that our own flags win.
- Duplicates across sources (a game in both Steam and Heroic) stay separate in v1; merging them is a backlog item.

Launch routes

Route	Used for	Permission tier	Playtime	From
Store URI through the OpenURI portal	Steam, Heroic, Lutris, Bottles	Base	Imported from the store	M1
Host spawn (argv list, no `--watch-bus`)	Desktop entries, Flatpak games, Faugus games, manual native games	Host opt-in	Own sessions (direct child), except Faugus (imported)	M2
umu inside the sandbox	Manual Windows games; manual native games in the Steam Linux Runtime	Proton	Own sessions	M3

1. Play picks the route from the game's launch kind; when the needed tier is missing, the game shows "needs permission" with instructions instead of failing.

2. Store launches save `last_played` and show a toast naming the launcher; "exit after launch" applies only to these.
3. Games we start get a unique `GAMEID` and window class, a Playing state with Stop, and sessions written every 30 s and on exit.
4. While a game we started is running, the app holds itself and asks the Background portal, so closing the window never ends the game.

Artwork

- Four layouts, as in OpenGamepadUI and SteamGridDB: portrait cover 2:3 (600×900), hero banner (about 1920×620), logo, icon.
- Priority: the source's local art, then SteamGridDB (user's own API key, stored with oo7), then a generated placeholder of the icon on a blurred background.
- Tile sizes Small 150×225, Medium 200×300 (default), Large 250×375. Art fills the tile when its width/height is within 0.667 ± 0.1 (Heroic's 3:4 art fills with a slight crop); square or landscape art fits on a blurred copy of itself.
- Users can replace any layout from file, URL or a SteamGridDB picker, Faugus-style.

Research: Research — Heroic Games Launcher · Research — Cartridges · Research — Faugus Launcher · Research — Flatpak & Flathub

Fullscreen gamepad mode

Fullscreen mode is a second, 10-foot presentation of the same models in the same window: top bars hidden, larger type, Home, Library and Game pages plus a menu drawer, all driven by a controller through libmanette. It hands games to their launchers and picks up again when the user comes back; it does not overlay running games or control hardware (OpenGamepadUI's session-level features are out of scope).

Entering and leaving

- Enter from the main menu, F11, the `--fullscreen` command-line flag, the Guide button while the desktop window is focused, or a "Start in fullscreen mode" preference.
- Leave with F11, Esc, or Exit fullscreen mode in the menu drawer; inside fullscreen mode the Guide button opens the drawer instead.
- The window calls `gtk_window_fullscreen()`, watches `fullscreened`, and sets `AdwToolbarView:reveal-top-bars` to false.

Screens

Screen	Content	Controller
Home	Hero banner of the focused game; carousel of recent and pinned games; a Library tile at the end	D-pad moves; A opens the game; Y search
Library	Tabs All, Installed, Favourites, then one per collection; 2:3 cover grid that remembers position per tab	LB/RB switch tabs; A opens; Y search; LT/RT cycle sort
Game	Hero, logo, large Play button with default focus, then Favourite, Add to collection, launch summary	A plays; B back; View opens options
Menu drawer (left)	Playing now card (Return, Stop) when we run the game; Home, Library, Collections, Settings subset, Exit fullscreen, Quit	Start or Guide toggles; B closes
Options sheet	Context actions for the focused game: Play, Favourite, Collections, Hide, Open in launcher	View or X opens; B closes
Search keyboard	Grid on-screen keyboard for the search field only	D-pad and A type; B deletes a character, or closes when the field is empty; Start confirms
Hint bar (bottom)	Hints for the focused item (for example A Select, B Back, Y Search) in our own neutral glyphs	–

Input model

- One gamepad service turns device events into app intents: navigate (up, down, left, right), activate, back, previous or next tab, search, menu, options, home.
- Navigate emits `move-focus` on the window (the arrow-key path); in grids it uses `gtk_grid_view_scroll_to` with focus and select flags. Intents never become synthetic key events.
- Held D-pad or stick repeats after 300 ms, then every 50 ms (Heroic's defaults), with a 0.5 stick deadzone.
- Glyphs and A/B order follow the device type (Xbox, PlayStation, Nintendo with A/B swapped, Steam Deck).

- Focus memory per page and tab; when focus is lost, the first focusable item on the page takes it; focus ring and a 1.05 scale on the focused tile; navigation sounds optional and off by default.
- Basic controller navigation also works in the desktop window, behind a preference.

Launch and return

1. Play shows an overlay that matches the route: "Opening Steam…" for store games, "Starting…" with hold-B cancel for games we run.
2. Store launches hand over and the app stays open in the background.
3. Games we start show a Playing state with Stop; the app keeps running for as long as they do.
4. When the window regains focus, it returns to the game's page with the Playing now card.
5. The Guide button cannot be caught while a game has focus; returning is done through the launcher, the system, or quitting the game.

Gamescope and handhelds

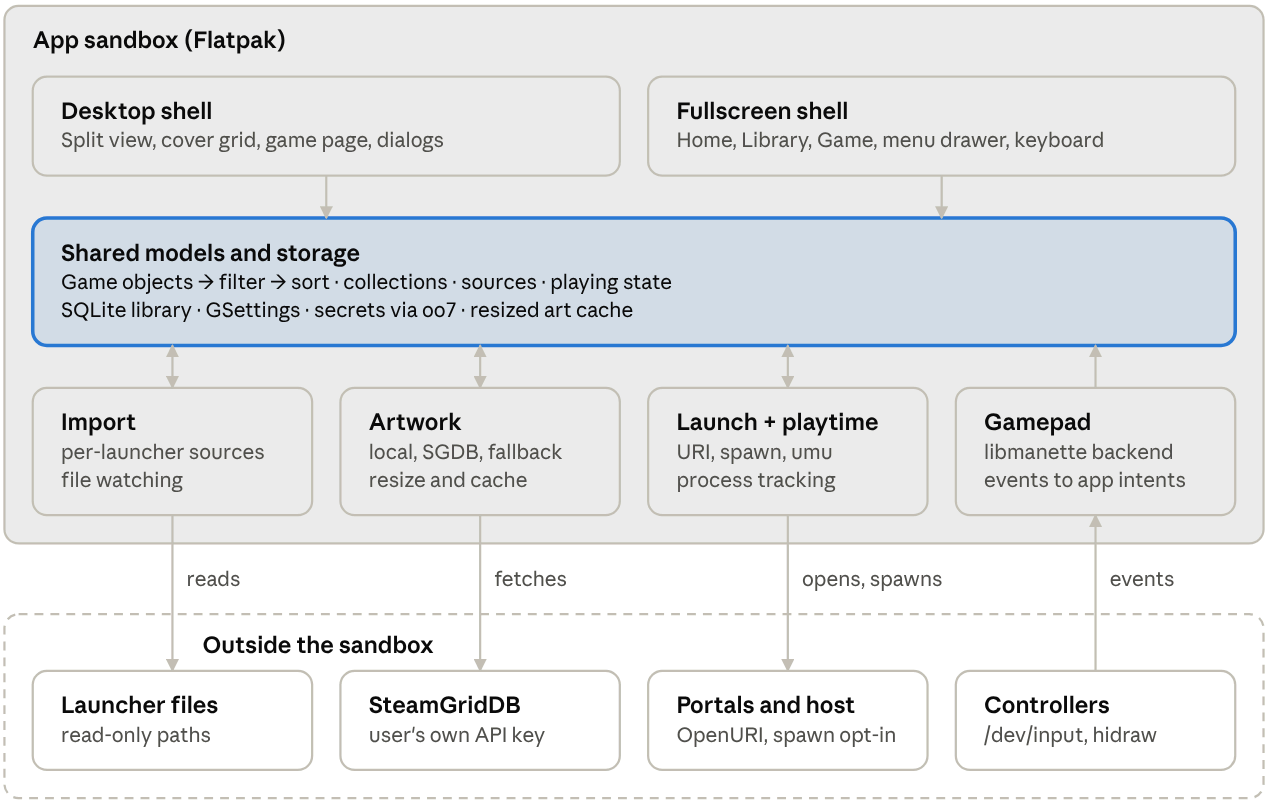
- Document running `gamescope -f -- flatpak run <app-id> --fullscreen` for couch setups; GTK runs as an X11 client there unless `--expose-wayland` is used.
- Gamescope focus atoms, overlays, TDP and controller remapping are out of scope for v1; a spike issue checks feasibility later.
- Layout targets: 1280×800 (Steam Deck, tested in Desktop Mode and in Game Mode as a non-Steam game) and 1920×1080 TV.
- MetaInfo declares `<supports><control>gamepad</control>`.

Research: Research — OpenGamepadUI · Research — GNOME Platform & libadwaita · Research — Heroic Games Launcher · Research — Xbox Game Library UI

Architecture, stack and data model

The app is Rust on GTK4 and libadwaita in two crates: a GTK-free core (sources, launching, playtime, storage) and a UI crate with two shells over the same GObject list models.

Both shells share one model; only services reach files, portals and the network



Architecture · two shells, shared models, four services, outside world

Shells never read files or spawn processes; they observe models and call services. That keeps fullscreen mode a pure presentation layer and lets the core be tested against fixture copies of launcher data.

Stack decisions

Area	Choice	Reason
Language	Rust: gtk4 0.11, libadwaita 0.9 with `v1_10`	Large grids, background scanning and image decoding need real threads; most active GNOME app community
UI definitions	Blueprint templates	In the GNOME SDK, editor support, used by Highscore and Cartridges
Build	Meson driving cargo; GNOME Builder template	Expected by GNOME and Flathub tooling

Area	Choice	Reason
Runtime	`org.gnome.Platform` 51 (libadwaita 1.10, GTK 4.24)	Gives `AdwSidebar` and current widgets
Gamepad	libmanette 0.2 behind a `GamepadBackend` trait; Rust bindings generated with `gir`	Swap to libmanette 1.0 or SDL3 later without touching UI
Storage	SQLite (rusqlite) with migrations	Thousands of games, sessions and collections; queries and atomic writes
Settings	GSettings	Window state and preferences, GNOME standard
Secrets	oo7 (Secret portal in Flatpak)	SteamGridDB key must not sit in plain settings
Portals	ashpd	OpenURI, file chooser, background
Images	glycin for decoding untrusted art; covers stored resized	Safe decoding, small memory per tile
Windows games	Bundled umu-launcher; Proton from `compatibilitytools.d` and our own tools folder	Same model as Lutris, Bottles, Faugus

Data model

Entity	Key fields
Game	`id` (`<source>:<native id>`), source, native ID, title, sort title, developer, platform (Windows or Linux), installed, install path, size, added at, last played at, favourite, hidden, pin order (up to three pinned), user-removed, missing since, launch kind (store URI, host command, desktop entry, Proton, runtime native), metadata (genres, description)
LaunchConfig (games we run)	Executable, arguments, working folder, environment, Proton version, prefix path, tools (MangoHud, GameMode), wrappers

Entity	Key fields
PlaySession	Game, started at, ended at, duration; own tracking only
ImportedPlaytime	Game, source, total, last played; used instead of sessions for sources that track playtime themselves (Steam, Heroic, Lutris, Faugus)
Artwork	Game, layout (cover 2:3, hero, logo, icon), origin (source, SteamGridDB, user, generated), file path
Collection	ID, name, order, member games
SourceState	Source, enabled, resolved location, user override, last sync, errors

Storage locations

- `$XDG_DATA_HOME/<app>/library.db`: SQLite in WAL mode, schema migrations from the first release.
- `$XDG_DATA_HOME/<app>/art/`: art the user chose; `$XDG_CACHE_HOME/<app>/art/`: fetched and generated art, safe to delete.
- A JSON export of collections, favourites, hidden games, manual games and own playtime, for backup (M5).

Metadata providers

1. The source's own data first: Steam `appinfo.vdf`, Heroic `GameInfo`, Lutris and Faugus fields.
2. SteamGridDB for grids (600×900), heroes, logos and icons, by Steam app ID or title search, cached and rate-limited.
3. Backlog: ProtonDB tier by Steam app ID, HowLongToBeat, PCGamingWiki (as Heroic shows them).

Testing: parser and resolver unit tests against fixture trees for native and Flatpak layouts of each launcher; UI smoke tests under xvfb in the Flatpak CI job.

Research: Research — GNOME Platform & libadwaita · Research — Cartridges · Research — OpenGamepadUI

Flatpak packaging and permissions

Ship on `org.gnome.Platform` 51 with permissions in three tiers: a small base set from M1, a Proton tier from M3, and host spawning as a user opt-in that the app detects at runtime.

Identity

- App ID `io.github.<owner>.<AppName>`, with dashes in the GitHub owner turned into underscores (for example `io.github.Tim_van_Oudheusden.<AppName>`). Fix the name and ID in M1 (#1), before any build is shared: a later rename moves users' data folder, and Flathub refuses most renames.
- Name of 15 characters or fewer (decided in #1, before the scaffold), an imperative summary of 10–25 characters, SVG icon, light and dark brand colours, `<supports> <control>gamepad</control>`.

Permission tiers

Tier	finish-args	Unlocks	From
Base	`wayland`, `fallback-x11`, `ipc`, `dri`, `network`; read-only data dirs of Steam (native and Flatpak), Heroic, Lutris, Bottles, Faugus; `xdg-data/applications:ro`; `host-os:ro` (for `/run/host/usr/share/applications`); `/var/lib/flatpak/exports:ro`, `xdg-data/flatpak/exports:ro`	Importing, artwork, store launches through OpenURI	M1 (#4)
Host (opt-in)	`--talk-name=org.freedesktop.Flatpak`, granted by the user with `flatpak override`	Desktop entries, Flatpak games, Faugus games and manual native games through host spawn	M2 (#23, #24)

Tier	finish-args	Unlocks	From
Proton	`x11` replaces `fallback-x11` (both together is a linter error; `x11` with `wayland` only a warning); `--device=all`; `--allow=multiarch,devel,per-app-dev-shm`; `pulseaudio`; `xdg-data/umu:create` (umu installs the Steam Runtime there); `~/Games:create`; Compat.i386, GL32 and MangoHud Vulkan-layer extensions; bundled umu-launcher	Windows games, and native games in the Steam Linux Runtime, inside the sandbox	M3 (#37)
Gamepad	`--device=input` paired with `--device-if=all:!has-input-device`; libmanette module if needed	Controller navigation (already covered by `--device=all` once the Proton tier ships)	M4 (#46)

- Every manifest change is its own `human-only` issue (#4, #37, #46, #63); feature issues never edit the manifest.
- Manual games are added by choosing the game's folder through the portal, because a game needs its whole folder, not one file. Steam libraries outside granted paths are read from the `apps` list in `libraryfolders.vdf`.
- GameMode goes through its portal (libgamemode 1.4+), never a `org.freedesktop.portal.*` talk-name.
- The app checks at start-up which tiers are present and explains a missing one in place (an `AdwBanner` or a row with a "How to enable" link), never as a launch failure.

Build and CI

- Rust SDK extension (branch 26.08) with offline cargo sources, or flatpak-pip-generator for Python, depending on the language decision.
- GitHub Actions with `flatpak/flatpak-github-actions/flatpak-builder@v6` in the `gnome-51` image; build, lint with `flatpak-builder-lint`, upload a bundle per PR.
- MetaInfo validated in CI: developer ID, launchable, captioned screenshots pinned to a tag, releases, OARS rating.

Flathub submission rules that shape the plan

- Generative-AI policy: the Flathub manifest must be written by a person with no AI assistance; AI-generated code, docs or packaging must be disclosed with its extent; Flathub PRs, descriptions and replies must not be written or opened by AI agents.

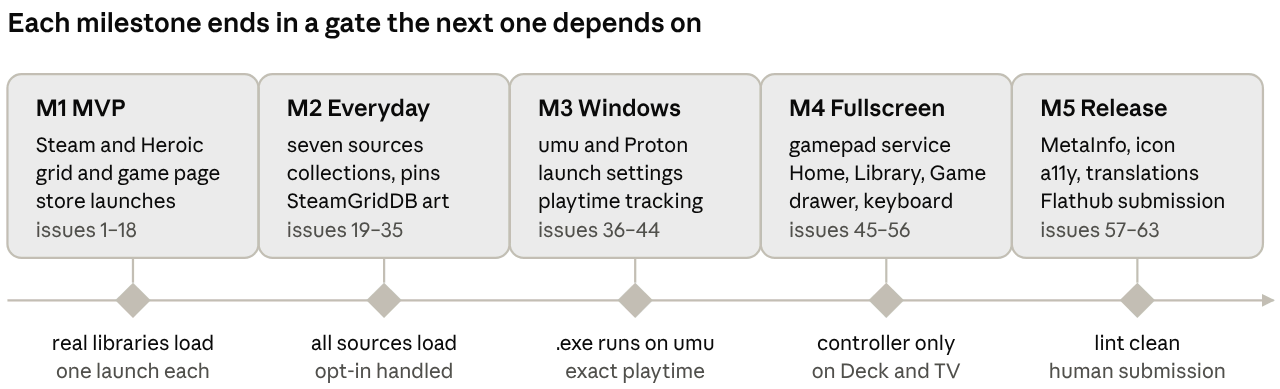
Issues generated from this spec count as AI-assisted planning, so the disclosure note goes into the repo with the issue import (#2) and is kept current.

- New apps need real development history and must offer substantial functionality beyond wrapping other launchers; submit after M4 with tagged releases, not at M1.
- Every permission must be explained in the submission PR; later builds that add permissions are held for moderation.

Research: Research — Flatpak & Flathub

Milestones, epics and GitHub issues

Five milestones (63 issues) take the app from a Steam-and-Heroic library to a Flathub release: M1 Library MVP, M2 Everyday library, M3 Windows games, M4 Fullscreen mode, M5 Release. Each milestone has a gate that must pass before the next starts; the tables below are the candidate issues, one row per GitHub issue.



Roadmap · five milestones and their gates (sequence only, not to scale)

Bands show order and scope, not dates; each diamond is the gate written at the top of that milestone's table.

Issue conventions

- Title in the imperative ("Import Steam libraries"); one milestone per issue; epics as GitHub milestones or a parent issue with a task list.
- Body: context with a link to the spec section, scope, acceptance criteria as a checklist, out of scope, research link.
- Labels: `area:core`, `area:import`, `area:ui`, `area:launch`, `area:art`, `area:gamepad`, `area:packaging`; `type:feature`, `type:spike`, `type:chore`; `human-only` for work Flathub requires a person to do (manifest, submission PR, review

replies).

- Spikes are time-boxed and end in a short write-up that updates this spec.

M1 Library MVP

Gate: a real Steam and Heroic library (native and Flatpak installs of each) imports, shows in the grid, and one game from each launches from the Flatpak build with base permissions only.

#	Issue	Labels	Acceptance criteria	Depends on
1	Decide name, app ID and licence	`area:packaging`, `type:chore`, `human-only`	Name of one or two plain nouns, 15 characters or fewer, no trademarks or platform names (no Xbox, GNOME, Adwaita); `io.github.<owner>.<Name>` fixed before any build is shared; LICENSE committed	–
2	Add the AI-use disclosure	`area:packaging`, `type:chore`, `human-only`	`AI_DISCLOSURE.md` committed with the issue import; names AI-assisted parts (research, spec, issues, code) and their rough extent; a release checklist item keeps it current	–
3	Scaffold Rust + Meson + Blueprint app on GNOME 51	`area:core`, `type:chore`	Builds with Meson; `AdwApplicationWindow` with About dialog; gettext wired for Rust and Blueprint; GSettings schema under the app ID; stub desktop file, MetaInfo and icon that pass the linter	1
4	Write the development Flatpak manifest by hand	`area:packaging`, `type:chore`, `human-only`	Base permission tier only; every finish-arg commented with its reason; the app runs from the Flatpak	3

#	Issue	Labels	Acceptance criteria	Depends on
5	Set up CI	`area:packaging`, `type:chore`	Flatpak build in the `gnome-51` image; blocking `flatpak-builder-lint`; clippy, rustfmt, unit tests; UI smoke test under xvfb; bundle uploaded per PR	3, 4
6	Core data model and SQLite store	`area:core`, `type:feature`	Tables Game (with platform, launch kind, favourite, hidden, pin order, user-removed, missing-since), LaunchConfig, Artwork, Collection and membership, PlaySession, ImportedPlaytime, SourceState; migrations; WAL; CRUD unit tests	3
7	Source location resolver	`area:import`, `type:feature`	Ordered native and Flatpak candidates with required sub-paths; resolved location and user override stored in SourceState; fixture tests	6
8	Import Steam libraries	`area:import`, `type:feature`	Every library in `libraryfolders.vdf`; install state from `appmanifest` files, or from the file's `apps` list for libraries outside granted paths; non-games filtered and platform set from `appinfo.vdf`; `librarycache` art; native and Flatpak Steam; fixture tests	7

#	Issue	Labels	Acceptance criteria	Depends on
9	Import Heroic games (Epic, GOG, Amazon, sideload)	`area:import`, `type:feature`	Library joined with each `installed.json`; `game-overrides.json` applied; platform from install info; hidden, favourites and categories imported once, after which our own flags win; credential files never opened; native and Flatpak Heroic	7
10	Sync, missing games and refresh	`area:import`, `type:feature`	Games gone from a source get `missing_since` and leave the views; reappearing games return with their user data; user-removed games stay removed while still in the source; refresh from the Add menu; progress in the header; errors collected in one summary dialog	8, 9
11	Library list models	`area:ui`, `type:feature`	Game objects → incremental filter → sort; with 5,000 fixture games, frames stay under 16 ms while scrolling and search updates within 100 ms on the reference machine named in the issue	6
12	Main window with hideable sidebar	`area:ui`, `type:feature`	`AdwOverlaySplitView` + `AdwSidebar` with Library and Sources sections and counts; header toggle and F9 show or hide it; overlay mode below 730 sp; usable at 360 sp	11

#	Issue	Labels	Acceptance criteria	Depends on
13	Cover tile widget	`area:ui`, `area:art`, `type:feature`	Sizes Small 150×225, Medium 200×300, Large 250×375; fill when width/height is within 0.667 ± 0.1, else fit on a blurred copy; icon placeholder; badges outside the bottom third; Play on hover and focus; accessible label with title, launcher, state	11
14	Tile context menu	`area:ui`, `type:feature`	Opens on right-click, long-press and Menu key; Play, Details, Open in launcher, Hide with undo; Delete acts only on the focused tile, with undo	13
15	Search, sort and filter row	`area:ui`, `type:feature`	Ctrl+F and type-to-search; case- and accent-insensitive matching on substrings and word prefixes, ignoring a leading "The"; five sort options; stacking toggles Installed, Not installed, Favourites, Windows, Native; status pages for empty and no results	12, 13
16	Game page	`area:ui`, `type:feature`	Hero with gradient and logo or title; Play has default focus; facts row; About group; ⋮ menu with Hide, Open in launcher, Open install folder; store games show a read-only launch summary and Open in Steam or Heroic; Alt+Left back	12

#	Issue	Labels	Acceptance criteria	Depends on
17	Launch store games through OpenURI	`area:launch`, `type:feature`	`steam://rungameid` and `heroic://launch?appName=…&runner=…` work in the Flatpak with base permissions; per-game option for `steam://launch/<id>/dialog`; last played saved; toast names the launcher; checked on GNOME that no app chooser appears	8, 9
18	Local artwork and art cache	`area:art`, `type:feature`	Source art used first; covers stored resized for each tile size; cache in `$XDG_CACHE_HOME`, safe to delete	8, 9

M2 Everyday library

Gate: all seven sources (Steam, Heroic, Lutris, Bottles, Faugus, desktop entries, manual) import; collections, favourites, pins, play history and SteamGridDB art work; with the host opt-in granted, desktop entries, Flatpak games, Faugus games and manual native games launch and record playtime, and without it they show "needs permission" instead of failing.

#	Issue	Labels	Acceptance criteria	Depends on
19	Import Lutris games	`area:import`, `type:feature`	`pga.db` copied then queried; `coverart/` art; playtime imported (column to verify); native and Flatpak Lutris; `lutris:rungameid` launch	7, 17
20	Import Bottles programs	`area:import`, `type:feature`	`library.yml` and custom bottle path read; `bottles:run` launch; grid art when present	7, 17

#	Issue	Labels	Acceptance criteria	Depends on
21	Import Faugus games	`area:import`, `type:feature`	`games.json` from the data dir (not the legacy config path); covers and banners; playtime imported from Faugus and not tracked by us; launch via `faugus-launcher --game <id>` or its Flatpak command on the host	7, 24
22	Import desktop entries and Flatpak games	`area:import`, `type:feature`	`Categories=Game`; NoDisplay and Hidden skipped; known launchers' own entries and their game shortcuts (`steam://`, `heroic://`, `lutris:`) skipped; reads `xdg-data/applications`, `/run/host/usr/share/applications` and the Flatpak exports dirs	7
23	Detect host-spawn permission and explain it	`area:launch`, `area:ui`, `type:feature`	Start-up check for `org.freedesktop.Flatpak`; affected games badged "needs permission"; a Preferences row with the exact `flatpak override` command	13, 22
24	Launch through host spawn	`area:launch`, `type:feature`	Argv lists only; desktop entries run from their parsed Exec line so the game is our child; Flatpak games via `flatpak run`; per-game environment with `--env`; no `--watch-bus`	23
25	Session tracker and Playing state	`area:launch`, `area:core`, `type:feature`	For every game we spawn: PlaySession written every 30 s and on exit; Playing badge and Stop; the app holds itself and asks the Background portal while a game runs; closing the window does not end the game; "exit after launch" applies to store launches only	24

#	Issue	Labels	Acceptance criteria	Depends on
26	Add and edit native games by hand	`area:ui`, `type:feature`	`AdwDialog`; game folder chosen through the portal (folder, not a single file); executable, title, arguments, working folder; cover, hero and logo pickers (file, URL); launches through host spawn	24
27	Collections	`area:ui`, `area:core`, `type:feature`	Create, rename, reorder, delete; add and remove from tile and game page; sidebar section (check `AdwSidebar` supports row menus and reordering, else a custom list); undo toasts	12, 14
28	Favourites, pins and Jump back in	`area:ui`, `type:feature`	Favourite toggle; up to three pins; row of six recent games, pins first, hidden during search and filters	15, 16
29	Play history and imported playtime	`area:ui`, `area:import`, `type:feature`	Page grouped by Today, This week, Earlier; imported totals from Steam (`localconfig.vdf` per account, to verify), Heroic `timestamp.json`, Faugus and Lutris; each source uses imported totals or our own sessions, never both; origin shown	9, 19, 21, 25
30	SteamGridDB artwork	`area:art`, `type:feature`	API key stored with oo7; picker per layout; one request at a time with backoff on HTTP 429; cached so it works offline after the first fetch	18
31	Watch sources for changes	`area:import`, `type:feature`	`Gio.FileMonitor` on each source's key files; resync debounced by 2 s and run off the main thread	10
32	Hidden games and removal	`area:ui`, `type:feature`	Ctrl+H shows hidden games; Unhide; Remove with undo sets user-removed	14

#	Issue	Labels	Acceptance criteria	Depends on
33	Preferences dialog	`area:ui`, `type:feature`	General (tile size, titles, badges, exit after launch), Sources (enable, location override, hide), Artwork (SteamGridDB key, prefer SteamGridDB)	7, 30
34	Keyboard shortcuts and help	`area:ui`, `type:feature`	`AdwShortcutsDialog` on Ctrl+? listing Ctrl+F, Ctrl+N, Ctrl+, , F9, Ctrl+H, Enter, Alt+Left, F11	12
35	GNOME Shell search provider and launch CLI	`area:core`, `type:feature`	`SearchProvider2` answers from the running app's model over D-Bus; `--launch <id>` works without opening the window	17

M3 Windows games

Gate: a standalone Windows game added from its .exe runs through bundled umu-launcher and a chosen Proton inside the Flatpak, with exact playtime, without the host opt-in.

#	Issue	Labels	Acceptance criteria	Depends on
36	Spike: umu and Proton inside our Flatpak	`area:launch`, `type:spike`	A test game runs; records the finish-args and extensions needed, the linter result for `x11` with `wayland`, the first-run Steam Runtime download, and whether games survive our main process exiting; findings written into this spec	4

#	Issue	Labels	Acceptance criteria	Depends on
37	Add the Proton tier to the manifest	`area:packaging`, `type:chore`, `human-only`	`x11` replaces `fallback-x11`; `--device=all`; multiarch, devel, per-app-dev-shm; pulseaudio; `xdg-data/umu:create`; Compat.i386, GL32 and MangoHud Vulkan-layer extensions; umu-launcher module; each line commented; lint passes or an exception is planned	36
38	Discover Proton builds and run umu	`area:launch`, `type:feature`	Proton found in native and Flatpak Steam `compatibilitytools.d` and our tools folder; first-run Steam Runtime download shows progress and a clear offline error	37
39	Download Proton builds (if decided)	`area:launch`, `area:ui`, `type:feature`	Preferences page lists GE-Proton and Proton-CachyOS releases with size and date; install and remove; never at launch time	38
40	Per-game launch settings	`area:launch`, `area:ui`, `type:feature`	Game page group: Proton version, prefix, arguments, environment, MangoHud, GameMode through its portal; free-text escape hatch; Windows games page in Preferences with default Proton, prefix folder and tools	16, 38

#	Issue	Labels	Acceptance criteria	Depends on
41	Add a Windows game from .exe or installer	`area:ui`, `type:feature`	Game folder through the portal; title and icon from exe metadata; "Run installer" then pick the installed executable; prefix under our games folder	26, 38
42	Track Proton games	`area:launch`, `type:feature`	Unique `GAMEID` and prefix per game; session tracker from #25 reused; Stop ends every process in the game's prefix	25, 38
43	Per-game logs	`area:launch`, `type:feature`	"Play with logs" keeps umu and Proton logs per game; open from the game page	42
44	Run native games in the Steam Linux Runtime	`area:launch`, `type:feature`	Native manual games run through umu's runtime without host spawn; host spawn only when the user chooses it	26, 38

M4 Fullscreen mode

Gate: on a 1080p TV and on a Steam Deck in both Desktop Mode and Game Mode (added as a non-Steam game, Steam Input set to gamepad), a user can enter fullscreen mode, find a game by tab or search, launch it and come back, using only a controller.

#	Issue	Labels	Acceptance criteria	Depends on
45	Spike: libmanette from Rust in the Flatpak	`area:gamepad`, `type:spike`	Bindings generated with `gir` for 0.2; events with `--device=input`; Steam Deck tested in Desktop Mode and Game Mode with Steam Input on and off; decision on bundling; manifest changes left to #46	3
46	Add gamepad access to the manifest	`area:packaging`, `type:chore`, `human-only`	`--device=input` paired with `--device-if=all:!has-input-device` (redundant once the Proton tier grants `--device=all`); libmanette module if #45 says so	45
47	Gamepad service and intents	`area:gamepad`, `type:feature`	`GamepadBackend` trait; hotplug; intents navigate, activate, back, tabs, search, menu, options, home; 300 ms then 50 ms repeat; 0.5 deadzone; device type for glyphs and A/B swap	46
48	Controller navigation in the desktop window	`area:gamepad`, `area:ui`, `type:feature`	Navigate emits `move-focus`; grid uses `scroll_to` with focus and select; A activates; B pops pages and closes dialogs; behind a preference	47

#	Issue	Labels	Acceptance criteria	Depends on
49	Fullscreen shell scaffold	`area:ui`, `type:feature`	Enter by menu, F11, `--fullscreen`, Guide while focused, or start-up preference; leave by F11, Esc or the drawer; bars hidden; larger type scale; focus memory per page and tab; Fullscreen page in Preferences (start in fullscreen, navigation sounds, off by default)	47
50	Fullscreen Home page	`area:ui`, `type:feature`	Hero banner of the focused game; recents carousel; Library tile; Y search	49
51	Fullscreen Library page	`area:ui`, `type:feature`	Tabs All, Installed, Favourites and one per collection; LB/RB switch; LT/RT cycle sort; grid position kept per tab	27, 49
52	Fullscreen Game page, launch overlay and return	`area:ui`, `area:launch`, `type:feature`	Play focused; overlay text matches the launch ("Opening Steam…" for store games, "Starting…" with hold-B cancel for games we run); Playing now card on return	25, 49

#	Issue	Labels	Acceptance criteria	Depends on
53	Menu drawer and options sheet	`area:ui`, `type:feature`	Drawer on Start or Guide with Playing now, Home, Library, Collections, Settings (tile size, sounds, exit after launch), Exit fullscreen, Quit; options sheet on View or X	49
54	On-screen keyboard for search	`area:ui`, `type:feature`	Letters, digits, space, backspace, confirm; D-pad and A type; B deletes a character and closes the keyboard when the field is empty; Start confirms	49
55	Context hint bar with controller glyphs	`area:ui`, `area:gamepad`, `type:feature`	Hints change with the focused page and widget; our own neutral glyph icons; controller brand names only in settings	47
56	Gamescope notes and focus-atom spike	`area:gamepad`, `type:spike`	README section for `gamescope -f -- flatpak run <app-id> --fullscreen`; libadwaita tested under gamescope with and without `--expose-wayland`; write-up on X11 focus atoms for v2	49

M5 Release

Gate: tagged releases with a development history behind them, a clean `flatpak-builder-lint` run, and a Flathub submission PR written and opened by a person.

#	Issue	Labels	Acceptance criteria	Depends on
57	Final icon and brand colours	`area:packaging`, `type:chore`	Icon on the 128 px template with symbolic and nightly variants; light and dark brand colours; no console brand marks	1
58	MetaInfo and screenshots	`area:packaging`, `type:chore`, `human-only`	Text written or reviewed by a person; developer ID, launchable, 3–6 captioned windowed screenshots pinned to a tag and using libre game art, releases, OARS, gamepad support tag; no console brand words; validates in CI	57
59	Accessibility pass	`area:ui`, `type:chore`	Orca announces every tile, badge and page title; every action reachable by keyboard; usable with High Contrast and Large Text; reduced motion turns off tile scaling and transitions	48, 49
60	Translations	`area:core`, `type:chore`	Weblate project connected to the gettext setup from #3	3
61	Performance pass	`area:core`, `area:ui`, `type:chore`	With 5,000 fixture games on the reference machine: interactive within 2 s of launch, frames under 16 ms while scrolling, search under 100 ms (proposed targets)	11, 29

#	Issue	Labels	Acceptance criteria	Depends on
62	Export and backup	`area:core`, `type:feature`	JSON export and import of collections, favourites, hidden games, manual games and own playtime	6, 25, 27
63	Submit to Flathub	`area:packaging`, `type:chore`, `human-only`	Manifest written by a person; PR to `flathub/flathub` `new-pr` explaining every permission; disclosure up to date; replies by a person	2, 4, 37, 46, 58

Backlog (unscheduled): list view next to the grid (HIG), merging duplicate games across sources into one entry with several launch options, ProtonDB and HowLongToBeat info, itch and standalone Legendary importers, selection mode for bulk actions, gamescope focus integration, InputPlumber support.

Risks, open decisions and research index

The biggest risks are Flathub acceptance (thin-wrapper, host-dependency and AI rules) and gamepad support in a sandboxed GTK app; both are handled by doing real work inside the sandbox and by spikes before commitments.

Risks

Risk	Impact	Mitigation
Flathub sees the app as a thin wrapper or host-dependent	Rejected submission	Own value: Windows games via umu (M3), fullscreen mode (M4); base tier works without host spawn; submit only after M4 with real history

Risk	Impact	Mitigation
Flathub and GNOME Circle AI policies	Rejected submission or review friction	`AI_DISCLOSURE.md` committed with the issue import (#2); manifest, Flathub PR and replies by a person (#4, #37, #46, #58, #63); maintainer can explain all code; Circle is closed to new apps anyway
Xbox trade dress or trademarks	Takedown, rejection	No Microsoft names, logos, green or button art; our own neutral glyph icons; controller brand names only in settings; described as inspired by console launchers
libmanette Rust bindings and the coming 1.0 API	Gamepad work stalls	`GamepadBackend` trait; spike #45 first; SDL3 as fallback
`--device=input` acceptance and Steam Deck needing hidraw	Gamepads missing for some users	`--device-if=all:!has-input-device` pairing; Proton tier already brings `--device=all`
Launcher data formats change (Heroic JSON, Steam binary `appinfo.vdf`)	Broken imports	Fixture tests per format; tolerant parsers; per-source error summary
No playtime for URI launches	Incomplete history	Import store-side totals and label their origin; own tracking for games we run
Thousands of covers	Slow scrolling, memory	`GtkGridView` recycling; resized art cache; decoding off the main thread
Scope for a small team	Milestones slip	Gates per milestone; backlog for everything not needed for the gate

Decisions for you

- [ ] App name, GitHub owner of the app ID, and licence (GPL-3.0-or-later fits the ecosystem); needed for #1 before any code.

- [ ] Confirm Rust and a fresh codebase; the issues assume it (forking Cartridges' Python rewrite would make the app GPL-3.0-or-later and inherit unfinished code).
- [ ] Proton downloads inside the app (#39) or only builds already in `compatibilitytools.d`.
- [ ] Follow the system colour scheme or prefer dark, as media-heavy apps may.
- [ ] Controllers to test against: Xbox, DualSense, Steam Deck, Nintendo.

Already settled in this spec, change here if you disagree: host spawn is opt-in; duplicates stay separate in v1; F11 toggles fullscreen mode; there is no list view in v1.

Open technical questions (answered inside issues): where Steam stores playtime (#29); whether emitting `move-focus` gives reliable spatial navigation in `GtkGridView` (#48); how libadwaita behaves under gamescope (#56); whether the OpenURI portal opens `heroic://` without a chooser on GNOME (#17); whether games started inside the sandbox survive our main process exiting (#36).

Research index

Doc	What it covers
Research — Xbox Game Library UI	Xbox PC app, console and Xbox mode libraries; tile and art rules; widget mapping
Research — Faugus Launcher	Faugus: umu and Proton launching, view modes, data files, CLI
Research — Heroic Games Launcher	Heroic: library UI, settings model, Console Mode, data files and `heroic://`
Research — Cartridges	Cartridges: importers, cover pipeline, GNOME UI, end-of-life status
Document	OpenGamepadUI: fullscreen screens, navigation layers, input model
Document	GNOME 51, libadwaita widgets, HIG rules, libmanette, language choice
Document	Flathub rules, permissions, launching from the sandbox, AI policy